SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I Praresh.R C191226039 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Understanding	20%	2	2	2	2	2
Method Selection	20%	2	2	2	2	2
Solution Process	25%	1	1	1	1	1
Accuracy of Calculation	20%	1	1	1	1	1
Interpretation of Results	15%	1	1	1	1	1

Total Marks 9/5 **/40**

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

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Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

ASSESSMENT TOOL 1 – COMPUTER NETWORKS (ANSWERS)

SIMATS ENGINEERING

ASSESSMENT TOOL 1 – COMPUTER NETWORKS

Course Code: CSA07

Name: Pranesh.R

Register Number: 1912260397

CO2: Configure IP addressing, subnetting and basic routing techniques.

Q1. University Network – 192.168.20.0/24

A /24 network has 256 total addresses. We assign the subnets consecutively from 192.168.20.0, using each subnet's required size.

Lab	Network address	Broadcast address	Valid host range	Usable hosts
Lab A (/26)	192.168.20.0	192.168.20.63	192.168.20.1 – 192.168.20.62	62
Lab B (/27)	192.168.20.64	192.168.20.95	192.168.20.65 – 192.168.20.94	30
Lab C (/28)	192.168.20.96	192.168.20.111	192.168.20.97 – 192.168.20.110	14

Method: /26 contains 64 addresses (64 − 2 = 62 usable); /27 contains 32 addresses (32 − 2 = 30 usable); /28 contains 16 addresses (16 − 2 = 14 usable).

Q2. Two Systems – /26 Subnet Mask

Given mask: 255.255.255.192 = /26. Each subnet has 64 addresses, so the last-octet subnet ranges increase by 64.

System	IP address	Network address	Broadcast address	Valid host range
System 1	192.168.1.130	192.168.1.128	192.168.1.191	192.168.1.129 – 192.168.1.190
System 2	192.168.1.190	192.168.1.128	192.168.1.191	192.168.1.129 – 192.168.1.190

Both IP addresses belong to the same subnet: 192.168.1.128/26. Both are valid host addresses because they are between 192.168.1.129 and 192.168.1.190.

Usable hosts per subnet = 2^(32 − 26) − 2 = 64 − 2 = 62.

Q3. Organization Network – 172.16.0.0/16

Assuming each department receives the next available subnet, allocation is as follows:

Department	Network address	Broadcast address	Valid host range	Usable hosts
HR (/25)	172.16.0.0	172.16.0.127	172.16.0.1 – 172.16.0.126	126
Sales (/26)	172.16.0.128	172.16.0.191	172.16.0.129 – 172.16.0.190	62
IT (/27)	172.16.0.192	172.16.0.223	172.16.0.193 – 172.16.0.222	30
Admin (/28)	172.16.0.224	172.16.0.239	172.16.0.225 – 172.16.0.238	14

Remaining unused range: 172.16.0.240 – 172.16.255.255, within the allocated block 172.16.0.0/16.

The unused range is based on sequential allocation starting from 172.16.0.0. The assigned subnets do not overlap.

Q4. Subnetting 172.16.10.0/24 into /27

1. Number of subnets: /27 borrows 3 bits from /24, so 2^3 = 8 subnets.

2. Usable hosts per subnet: 2^(32 − 27) − 2 = 32 − 2 = 30 hosts.

3. A /27 subnet has blocks of 32 addresses. The address 172.16.10.78 falls in the range 172.16.10.64–172.16.10.95.

4. Network address: 172.16.10.64. Broadcast address: 172.16.10.95. Valid host range: 172.16.10.65–172.16.10.94.

5. The address 172.16.10.95 is the broadcast address of this subnet, so it is not a valid host address. It is within the same subnet block, but cannot be assigned to a computer.

Q5. IPv6 Address – /64

Given: 2001:0db8:0000:0000:0000:ff00:0042:8329

1. Shortest compressed form: 2001:db8::ff00:42:8329

2. Expanded full form: 2001:0db8:0000:0000:0000:ff00:0042:8329

3. Network prefix with /64: 2001:0db8:0000:0000::/64 (also written 2001:db8:0:0::/64).

4. Number of host bits: 128 − 64 = 64.

5. Total possible addresses: 2^64 = 18,446,744,073,709,551,616.

Note: IPv6 does not use an IPv4-style broadcast address; it uses multicast.

Short Conclusion

Subnetting divides a large IP network into smaller networks. Correct subnet masks help identify network addresses, broadcast addresses and valid host ranges. IPv6 compression makes addresses shorter to write, while the /64 prefix identifies the network portion.